

DMS Network — AWS Cost Calculator

Quick Cost Reference

Scenario 1: Demo/POC (Recommended — This Guide)

Single t3.micro EC2 + Docker Compose + RDS micro + S3

YEAR 1 (Free Tier):

EC2 t3.micro	\$0	✓
RDS db.t3.micro	\$0	✓
S3 (5GB free tier)	\$0	✓
Lambda (1M invokes free)	\$0	✓
Data transfer (first 1GB)	\$0	✓
Overages (100GB @ \$0.09/GB)	~\$5	

TOTAL/MONTH: \$5–10 💰

YEAR 2+ (Post Free-Tier):

EC2 t3.micro	\$12
EBS 20GB gp3	\$2

Cost Breakdown by Service

Service	Demo	Staging	Production
Compute	\$12 (t3.micro)	\$25 (2×t3.small)	\$390 (EKS + 3×t3.large)
Database	\$0–50	\$0–80	\$500
Caching	\$0	\$24–72	\$150
Event Bus	\$0 (embedded)	\$0 (SQS)	\$400 (MSK)
Search	\$0 (Postgres FTS)	\$0 (DynamoDB)	\$300 (OpenSearch)
Storage	\$0 (S3 free)	\$1	\$50
Network	\$0 (Elastic IP)	\$20	\$180
Observability	\$0–5	\$5–10	\$100
Total/Month	\$12–22	\$50–180	\$2,000+

Free Tier Timeline

✓ AWS Free Tier Eligibility (12 Months)

SERVICE	DEMO USAGE	FREE TIER	COST AFTER YR 1
EC2 t3.micro	✓ 750h/mo	✓ 750h/mo	\$0.0104/hr → \$12/mo
EBS	✓ 20GB	✓ 30GB	\$0.10/GB → \$2/mo
RDS db.t3.micro	✓ 1GB	✓ 750h/mo	\$0.018/hr → \$50/mo
S3	✓ 5GB	✓ 5GB + 20K	\$0.023/GB
Lambda	✓ 1M calls	✓ 1M/mo	\$0.20 per 1M after
DynamoDB	✓ scan	✓ 25 RCU/WCU	25 = \$1.25/mo (perpetual)
CloudWatch	✓ logs	✓ 5GB/mo	\$0.50/GB after
SQS	✓ events	✓ 1M/mo free	\$0.50 per 1M after
SNS	✓ email	✓ 1M/mo free	\$0.50 per 1M after
Data Transfer Out	✓ first 1GB	✓ 1GB/mo	\$0.09/GB
TOTAL (Year 1):		~\$5–15/month	
TOTAL (Year 2+):		~\$25–35/month	

Upgrade Path (as Usage Grows)

WEEK 1–4: Demo on t3.micro

- 1 instance, 1GB RAM
- All features work
- Cost: \$5–15/mo

MONTH 2–3: Add High Availability

- Upgrade to 2 × t3.small
- Add RDS db.t3.small backup
- Enable multi-AZ
- Cost: \$50–80/mo

MONTH 4–6: Prepare for Scale

- Add ElastiCache (Redis cluster)
- Switch to ECS on t3.medium
- Enable CloudWatch X-Ray tracing
- Cost: \$150–200/mo

MONTH 7+: Production Ready

- Migrate to EKS (3 nodes)
- Add MSK Kafka (event bus)
- Add OpenSearch (full-text search)
- Enable WAF + CloudFront CDN
- Cost: \$1,500–2,000/mo

Each step is backward compatible – NO CODE CHANGES.

Money-Saving Tips

1. Use Reserved Instances (save 30–50%)

```
Instead of on-demand:  
t3.small on-demand:    $0.0208/hr = $152/mo  
t3.small 1-yr RI:      $0.0126/hr = $92/mo  
→ Saves $60/month
```

2. Turn Off When Not Demoing

```
# Stop instance (keep data, pay for EBS only)  
aws ec2 stop-instances --instance-ids i-xxx  
  
# Savings: $12/mo (compute) but keep $2/mo (storage)  
# Perfect for dev/staging!
```

3. Use S3 Transfer Acceleration (for large uploads)

Example: Scaling Scenarios

Scenario A: "Ship POC in 2 weeks"

Initial: t3.micro (\$0)
Demo phase: +load test → t3.small (\$12/mo)
Customer review: 3 months @ \$12–25/mo
Budget: $\$25 \times 3 = \75

Scenario B: "Scale to 10 concurrent users"

Initial: t3.micro (\$0)
Month 3: Add t3.small + RDS micro (\$50/mo)
Month 6: Upgrade to t3.large × 2 + RDS small (\$150/mo)
Months 1–6: $\$0 + \$0 + \$0 + \$50 + \$50 + \$150 = \$250$

Scenario C: "Full enterprise deployment"

Dev: t3.micro ($\$0 \times 3$ months)

Hidden Costs to Avoid

Hidden Cost	Annual	How to Avoid
Data Transfer Out (DXO)	\$1,080 (12TB @ \$0.09/GB)	Use CloudFront (\$0.085/GB) or keep data in same region
NAT Gateway	\$384 (2 × \$32 × 6 months)	Use Elastic IP (free!) or VPC endpoints
Unused Elastic IPs	\$360 (12 × \$3/mo unattached)	Deallocate or attach to instance
Leftover EBS snapshots	\$48 (10GB snapshot retention)	Delete old snapshots, use lifecycle policies
CloudWatch Log Groups (verbose)	\$600 (10GB/mo @ \$0.50/GB)	Use log retention, aggregate to S3

Billing Alerts (Set These Up Now!)

```
# CloudWatch Alarm: Alert if > $50/month
aws cloudwatch put-metric-alarm \
  --alarm-name DMS-Cost-Alert-50 \
  --alarm-description "Alert if DMS costs exceed $50/month" \
  --metric-name EstimatedCharges \
  --namespace AWS/Billing \
  --statistic Maximum \
  --period 3600 \
  --threshold 50 \
  --comparison-operator GreaterThanThreshold \
  --alarm-actions arn:aws:sns:us-east-1:xxx:cost-alerts
```


Summary

Duration	Scenario	Setup Cost	Monthly	Total
2 weeks	Demo PoC	Free	\$5	\$5
3 months	Staging	Free	\$25	\$75
6 months	Scale	Free	\$100	\$600
12 months	Year-1 Production	Free	\$2,000	\$24,000

Bottom line: Start at **\$5–10/month**, scale to **\$100–500/month**, and reach production at **\$1,500–3,000/month** only when you're ready.